

The Impact of Noise on SNR in EEG Signals

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Abstract—This paper analyses the impact of additive noise on the Signal-to-Noise Ratio in EEG signals. Also investigated is the effectiveness of different filters in removing the noise, and the impact on power analysis in the different EEG signal bands.

Index Terms—EEG Signals, Signal Processing, IIR Filters, Additive Noise, Power Analysis

I. INTRODUCTION

An EEG is an important tool for monitoring brain activity and diagnosing brain conditions such as epilepsy, tumours, encephalopathy, stroke, and many others. Removal of common sources of noise in the EEG signal is a key step in many EEG signal processing pipelines to ensure the signal provides useful diagnostic information. Noise in EEG signals can come from a variety of sources, including eye movement and blinking (EOG noise), cardiac signals near the neck (ECG noise), other muscle activity such as jaw movement (EMG noise), electrode movement e.g. due to heartbeats (BCG noise), baseline wander (e.g. due to sweat), mains hum, and more. This paper investigates how an artificial additive noise source (consisting of broadband background noise, a 20Hz narrowband sinusoid, and simulated eye blinking) effects the SNR of the EEG signal. Also investigated is the effectiveness of different filters in removing the noise, and the impact on Power Spectral Density, as well as individual the individual EEG signal band powers:

- Gamma (35+ Hz): Heightened Awareness
- Beta (13 - 35 Hz): Problem Solved
- Alpha (8 - 13 Hz): Relaxed Thought
- Theta (4 - 8 Hz): Meditation / Creativity
- Delta (0.5 - 4 Hz): Deep Sleep

II. TEST CONDITIONS

A clean EEG signal and an artificial noise signal were loaded from the provided MAT file (sourced from the PhysioNet archive), sampled at 160 Hz. The noise signal contains three components typically found in EEG recordings: broadband background noise, 20 Hz narrowband sinusoidal interference (designed to mimic mains hum, but at an easier frequency to deal with for this assignment), and eye-blink artefacts. The noise was added to the clean EEG at a range of SNR levels: 40 dB, then from 5 dB to -25 dB in steps of -5 dB. The noise signal was scaled to achieve each target SNR using the average power over the entire signal length:

$$K = \sqrt{\frac{P_{signal}}{P_{noise} \cdot 10^{\frac{SNR}{10}}}} \quad (1)$$

where P_{signal} and P_{noise} are the powers of the clean EEG and raw noise respectively, and K is the scaling factor applied to the noise before addition.

Each noisy signal was then filtered using a two-stage cascade filter: a 4th-order Butterworth bandpass filter (0.5–40 Hz), followed by a second-order IIR notch filter at 20 Hz. The notch filter was implemented with zeros on the unit circle and poles at radius r , both at the notch frequency:

$$\theta_0 = 2\pi \cdot \frac{f_{notch}}{f_s} = \frac{\pi}{4} \quad (2)$$

This gives the filter's transfer function as:

$$H(z) = \frac{1 - 2\cos(\frac{\pi}{4})z^{-1} + z^{-2}}{1 - 2r\cos(\frac{\pi}{4})z^{-1} + r^2z^{-2}} \quad (3)$$

Three values of pole radius were used: $r = 0.99$, $r = 0.96$, and $r = 0.93$, giving three filter combinations in total. All filtering was applied using `filtfilt` for zero-phase operation. For each SNR level and filter combination, the SNR after filtering was estimated using the residual noise (i.e. the difference between the filtered noisy signal and the filtered clean signal). Spectral analysis was carried out using Welch's Method (`pwelch` in GNU Octave), and band power was calculated in each of the four primary EEG bands: delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz). The geometric mean of the four band powers was also computed as a summary measure of spectral shape.

III. RESULTS

Firstly, looking at the band powers *after* applying the noise but *before* filtering - expressing these in dB with the clean signal as the reference at 0 dB makes it quite clear what effect the noise is having on power. Given the raw power values span an enormous range, it makes sense to express the power as a dB difference for the sake of comparison. Taking the difference from the clean EEG signal, at 0 dB the measured power exactly equals the reference signal, i.e. there is no distortion of the signal. For small values of less than 1 dB, there is a distortion but it is negligible, and by 3 dB the noise will have doubled the apparent power of the signal, and at that point, the distortion will have undoubtedly rendered the signal useless.

Table I clearly shows that the Beta band is the most severely affected by the noise. This is expected, given the noise contains a 20 Hz narrowband sinusoid, which sits squarely within the Beta band. By the time the SNR decreases to just 5 dB, the

TABLE I
BAND POWERS BEFORE FILTERING (DB DIFFERENCE FROM CLEAN)

SNR Before	Delta	Theta	Alpha	Beta	Geo Mean
Clean	0.00	0.00	0.00	0.00	0.00
40	0.00	0.00	0.00	0.01	0.00
5	0.32	0.19	0.37	6.23	1.78
0	1.05	0.57	0.92	10.42	3.24
-5	2.84	1.57	2.22	15.12	5.44
-10	6.05	3.73	4.74	20.01	8.63
-15	10.31	7.22	8.46	24.98	12.74
-20	15.07	11.60	12.92	29.96	17.39
-25	20.01	16.38	17.71	34.96	22.26

beta band has already been severely overwhelmed with noise, with the Delta, Theta, and Alpha bands starting to show the noise creeping in. The Delta band is the “next to go” along with the Alpha band. The Delta band is most affected by the broadband noise and eye-blink artefacts which occur at a very low frequency. No narrow-band interference falls in the Alpha band allowing it slightly more resistance to the noise than the Delta band, however by -10 dB the broadband noise is strong enough to destroy the signal.

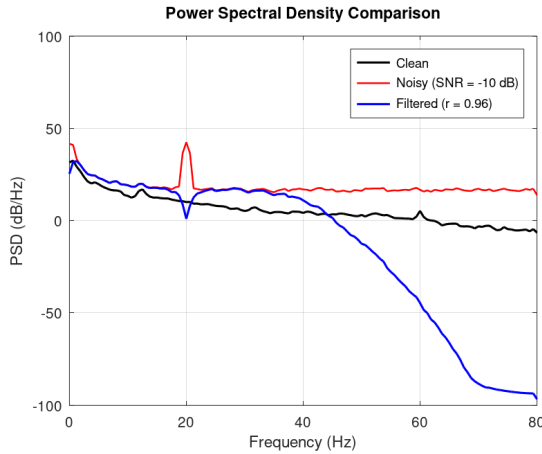


Fig. 1. Power Spectral Density plot comparing clean EEG signal, noisy EEG signal (SNR = -10 dB) and filtered ECG Signal ($r = 0.96$)

The Power Spectral Density plot of the clean, noisy, and filtered signals demonstrate the makeup of the noise very clearly, as well as the effects of the two-stage filter on the signal. The eye blink artefacts in can be seen in the Delta band in the noisy signal, and the 20 Hz sinusoidal noise shows as a peak in the signal at 20 Hz. The broad band noise also presents itself, as a near-fixed increase in the power density across the spectrum. The effects of the notch filter and IIR filter are also clear, significantly decreasing the power around 20 Hz and cutting off the signal above 40 Hz. Notably, the Butterworth filter’s lower cut-off of 0.5 Hz provides strong compensation for the eye-blink artefacts.

Table II shows that all three of the two-stage filters consistently improve SNR by $\bar{7}$ dB at all noise levels. Although the $r = 0.93$ filter provides the highest gain at 7.30 dB, the margin

TABLE II
SNR BEFORE AND AFTER FILTERING (DB)

SNR	$r = 0.99$	$r = 0.96$	$r = 0.93$
40	47.13	47.23	47.30
5	12.13	12.23	12.30
0	7.13	7.23	7.30
-5	2.13	2.23	2.30
-10	-2.87	-2.77	-2.70
-15	-7.87	-7.77	-7.70
-20	-12.87	-12.77	-12.70
-25	-17.87	-17.77	-17.70

over $r = 0.99$ is only 0.17 dB, showing that all three filters perform similarly. This is because the dominant source of SNR improvement in the two-stage filter is from the bandpass stage, which is rejecting the out-of-band broadband noise (seen on the right side of Figure 1), rather than from the notch filters removing the 20 Hz sinusoid. The Notch Filter only needs to operate on a narrow portion of the spectrum, thus widening it has minimal impact, as shown.

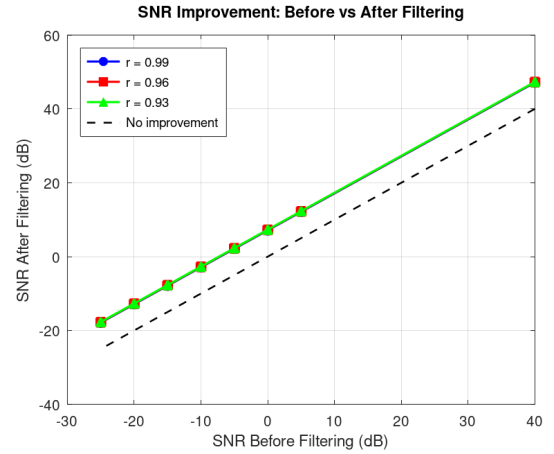


Fig. 2. Comparison of SNRs before and after filtering using 3 different notch filters

The plot of SNRs before and after filtering further demonstrates this, with all the lines drawn by all three filters practically overlapping at the same distance above No Improvement. Figure 3 redraws the Y-axis to show the SNR gain (i.e. the difference between SNR before and SNR after). This shows two key characteristics of the gain response, firstly that each of the filters are within about 0.1 dB of their neighbour, so there’s really no improvement. Secondly, the gain is constant regardless of input SNR, which follows from the fact that the filter is an LTI filter, so it has a fixed frequency response. As a result, the filter doesn’t care about the amount of noise, it removes the same proportion regardless. Thus the linear SNR gain is a property of the filter, explaining why the linear response in frequency results in a linear response in gain.

Tables III, IV, and V show the band powers after filtering, for each pole radius tested. Comparing these values to Table I, there is a significant improvement in the Beta band, where

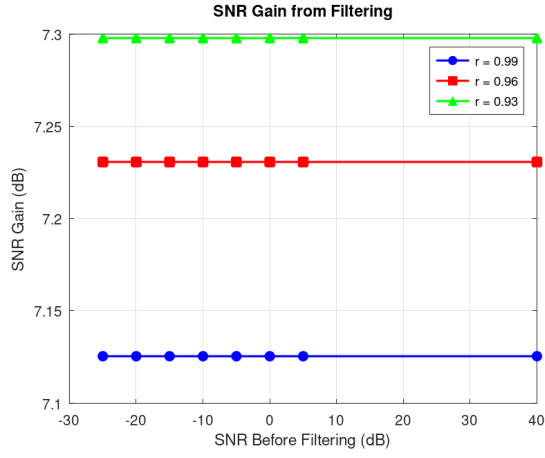


Fig. 3. Plot of SNR Gain (dB difference) of each notch filter

TABLE III
BAND POWERS AFTER FILTERING ($r = 0.99$)

SNR Before	Delta	Theta	Alpha	Beta	Geo Mean
Clean	0.00	0.00	0.00	0.00	0.00
40	-1.35	-0.00	-0.00	-0.34	-0.42
5	-1.25	0.19	0.36	0.32	-0.09
0	-0.92	0.57	0.91	1.37	0.48
-5	0.08	1.56	2.22	3.56	1.85
-10	2.34	3.72	4.73	7.04	4.46
-15	5.99	7.22	8.46	11.40	8.26
-20	10.50	11.60	12.92	16.15	12.79
-25	15.36	16.38	17.71	21.06	17.63

at an SNR of -10 dB, Beta's power drops from 20.01 dB above baseline to just 7.04 dB, 5.99 dB, and 5 dB for $r = 0.99, 0.96, 0.93$ respectively. This shows that the notch filter is doing the heavy lifting in removing the 20 Hz interference from this band. This does come at a cost though, which is observed in the 40 dB SNR case, where the Beta frequency range shows power gains of -0.34, -1.34, and -2.39 dB for $r = 0.99, 0.96, 0.93$. These negative values show that the notch filter is actually attenuating legitimate EEG signal content around 20 Hz, with the attenuation becoming worse as the pole radius decreases (i.e., as the notch width expands). With a power loss of only -0.34 dB at 40 dB SNR for $r=0.99$ and a gain of 7.125 dB SNR in filtering, the filter with pole radius 0.99 presents itself as the best choice for eliminating noise while preserving data.

The -1.35 dB power loss in the Delta band is consistent across all three filter combinations indicates that the Delta band attenuation comes from the bandpass filter's fall-off near its 0.5 Hz lower cut-off frequency rather than the notch filter. Delta is generally the most difficult band to deal with, where at an SNR of -25 dB it shows a power increase of 15.36 dB regardless of the pole radius. This is because the broadband noise and eye-blink noise which fall within the 0.5-4 Hz band aren't well covered by the Butterworth filter nor the notch filter.

TABLE IV
BAND POWERS AFTER FILTERING ($r = 0.96$)

SNR Before	Delta	Theta	Alpha	Beta	Geo Mean
Clean	0.00	0.00	0.00	0.00	0.00
40	-1.35	-0.01	-0.07	-1.34	-0.69
5	-1.25	0.18	0.29	-0.71	-0.37
0	-0.92	0.56	0.85	0.32	0.20
-5	0.08	1.55	2.15	2.50	1.57
-10	2.33	3.71	4.67	5.99	4.18
-15	5.99	7.20	8.39	10.35	7.98
-20	10.50	11.58	12.86	15.12	12.52
-25	15.36	16.37	17.65	20.04	17.35

TABLE V
BAND POWERS AFTER FILTERING ($r = 0.93$)

SNR Before	Delta	Theta	Alpha	Beta	Geo Mean
Clean	0.00	0.00	0.00	0.00	0.00
40	-1.35	-0.04	-0.22	-2.39	-1.00
5	-1.26	0.15	0.15	-1.76	-0.68
0	-0.92	0.53	0.70	-0.72	-0.10
-5	0.07	1.52	2.01	1.49	1.27
-10	2.33	3.68	4.53	5.00	3.89
-15	5.98	7.17	8.26	9.38	7.70
-20	10.50	11.56	12.72	14.15	12.23
-25	15.36	16.34	17.51	19.07	17.07

The geometric mean provides a useful single-number summary of the distortion to the shape of the power spectral density graph in comparison to the clean signal. When the noise inflates the power in a particular band more than others, such as the 20 Hz sinusoid in the Beta band, the geometric mean will move, showing that the ratio of power between bands has changed. At -10 dB SNR, the geometric mean drops from 8.63 dB in the unfiltered case to 4.46, 4.18, and 3.89 dB for the respective values of r . The geometric mean indicates that the wider notches are better at maintaining the original power ratios between the bands, but even at $r = 0.93$ there is still nearly 4 dB of distortion at -10 dB SNR, and it comes at the expense of more distortion in better signal conditions. An analysis of the consistency of real-world noise would be needed to truly say which filter is best, but for this experiment $r = 0.99$ seems to be the best at scrubbing the noise without distorting the underlying signal too severely.

IV. DISCUSSION & CONCLUSION

The results of this study show that noise affects EEG power bands unevenly depending on the types of noise present. In this study, the Beta band was most severely effected due to a sinusoid mimicking mains hum at a lower frequency. The Delta band was second most affected due to the broadband noise combined with the low-frequency eye-blink artefacts. A two-stage filtering solution proved to be useful in improving SNR, but the majority of the improvement came from rejecting broadband noise rather than specific types of noise within the signal. Notch filter pole radius had minimal impact on SNR overall, but was very useful in removing the 20 Hz sinusoid

present in the signal, however this was not absolute and there was attenuation of the underlying EEG signal. Overall, no combination of bandpass and notch filter could compensate for the broadband noise that fell within the EEG signal's own frequency bands (0.5 - 40 Hz in this case), which highlights the need for something more advanced such as an adaptive filter when dealing with real-world EEG signals.